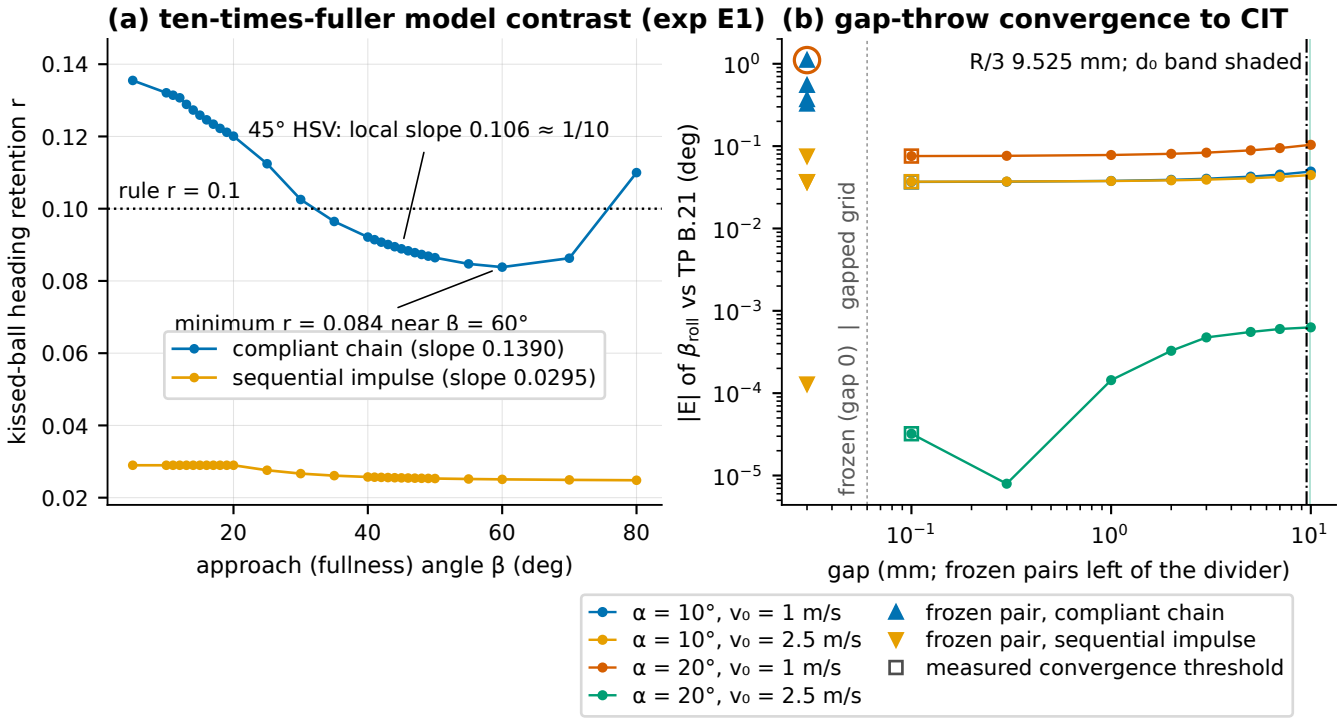
(a) ten-times-fuller model contrast (exp E1)
(b) gap-throw convergence to CIT
kissed-ball heading retention r
0.14
0.12
0.10
0.08
0.06
0.04
0.02
45° HSV: local slope 0.106 ≈ 1/10
rule r = 0.1
minimum r = 0.084 near β = 60°
compliant chain (slope 0.1390)
sequential impulse (slope 0.0295)
20
40
60
80
approach (fullness) angle β (deg)
|E| of β_{roll} vs TP B.21 (deg)
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
R/3 9.525 mm; d₀ band shaded
frozen (gap 0) | gapped grid
10^{-1}
10^{0}
10^{1}
gap (mm; frozen pairs left of the divider)
α = 10°, v₀ = 1 m/s
α = 10°, v₀ = 2.5 m/s
α = 20°, v₀ = 1 m/s
α = 20°, v₀ = 2.5 m/s
frozen pair, compliant chain
frozen pair, sequential impulse
measured convergence threshold